CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 4: Expansions

Topic-wise short notes with solved examples

The standard algebraic identities for squares and cubes of binomials and trinomials, the relations between sums, differences and products, and their use in finding values without long multiplication.

1 Squares of binomials

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)(a-b) = a^2 - b^2$
- $(x+a)(x+b) = x^2 + (a+b)x + ab$

Example 1.1 Expand $(3x+4y)^2$ and $\left(2a - \frac{1}{2a}\right)^2$.

SOLUTION

$(3x)^2 + 2(3x)(4y) + (4y)^2$ and $(2a)^2 - 2(2a)\left(\frac{1}{2a}\right) + \frac{1}{4a^2}$

$9x^2 + 24xy + 16y^2$ **and** $4a^2 - 2 + \frac{1}{4a^2}$

Example 1.2 Evaluate 103×97 and $(998)^2$.

SOLUTION

$103 \times 97 = 100^2 - 3^2 = 9991$

$998^2 = (1000-2)^2 = 1000000 - 4000 + 4$

9991 and 996004

2 Relations between sum, difference and product

- $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$
- $(a+b)^2 - (a-b)^2 = 4ab$
- $a^2 + b^2 = (a+b)^2 - 2ab = (a-b)^2 + 2ab$
- $(a+b)^2 = (a-b)^2 + 4ab$

Example 2.1 If $a+b=7$ and $ab=10$, find a^2+b^2 and $a-b$.

SOLUTION

$a^2+b^2=49-20=29$

$(a-b)^2=49-40=9$

$a^2+b^2=29;\ a-b=\pm 3$

Example 2.2 If $a-b=4$ and $a^2+b^2=40$, find ab.

SOLUTION

$16=40-2ab$

$ab=12$

3 Expressions with a number and its reciprocal

- $\left(a+\frac{1}{a}\right)^2=a^2+\frac{1}{a^2}+2$, so $a^2+\frac{1}{a^2}=\left(a+\frac{1}{a}\right)^2-2$.
- $\left(a-\frac{1}{a}\right)^2=a^2+\frac{1}{a^2}-2$, so $a^2+\frac{1}{a^2}=\left(a-\frac{1}{a}\right)^2+2$.
- $\left(a+\frac{1}{a}\right)^2-\left(a-\frac{1}{a}\right)^2=4$.
- $a^4+\frac{1}{a^4}=\left(a^2+\frac{1}{a^2}\right)^2-2$.

Example 3.1 If $a+\frac{1}{a}=5$, find $a^2+\frac{1}{a^2}$ and $a^4+\frac{1}{a^4}$.

SOLUTION

$a^2+\frac{1}{a^2}=25-2=23$

$a^4+\frac{1}{a^4}=23^2-2$

23 and 527

Example 3.2 If $a^2-3a+1=0$, find $a+\frac{1}{a}$ and $a^2+\frac{1}{a^2}$.

SOLUTION

Divide by a: $a-3+\frac{1}{a}=0$, so $a+\frac{1}{a}=3$.

3 and 7

Example 3.3 If $a - \frac{1}{a} = 4$, find $a + \frac{1}{a}$.

SOLUTION

$\left(a + \frac{1}{a}\right)^2 = 16 + 4 = 20$

$\pm 2\sqrt{5}$

4 Square of a trinomial

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

- For a minus sign, treat the term as negative: $(a - b + c)^2 = a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$.
- So $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca)$.

Example 4.1 Expand $(2x - 3y + z)^2$.

SOLUTION

$4x^2 + 9y^2 + z^2 - 12xy - 6yz + 4zx$

Example 4.2 If $a + b + c = 10$ and $a^2 + b^2 + c^2 = 38$, find $ab + bc + ca$.

SOLUTION

$100 = 38 + 2(ab + bc + ca)$

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5 Cubes of binomials

- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$
- So $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$ and $a^3 - b^3 = (a - b)^3 + 3ab(a - b)$.
- $a^3 + \frac{1}{a^3} = \left(a + \frac{1}{a}\right)^3 - 3\left(a + \frac{1}{a}\right)$ and $a^3 - \frac{1}{a^3} = \left(a - \frac{1}{a}\right)^3 + 3\left(a - \frac{1}{a}\right)$.

Example 5.1 Expand $(2x - 3y)^3$.

SOLUTION

$(2x)^3 - 3(2x)^2(3y) + 3(2x)(3y)^2 - (3y)^3$

$8x^3 - 36x^2y + 54xy^2 - 27y^3$

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Example 5.2 If $a+b=5$ and $ab=6$, find a^3+b^3.

SOLUTION

$125-3\times 6\times 5$

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Example 5.3 If $x+\frac{1}{x}=4$, find $x^3+\frac{1}{x^3}$.

SOLUTION

$4^3-3\times 4$

52

Example 5.4 Evaluate $(101)^3$.

SOLUTION

$(100+1)^3=1000000+30000+300+1$

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6 Sum and product identities

- $(a+b)(a^2-ab+b^2)=a^3+b^3$
- $(a-b)(a^2+ab+b^2)=a^3-b^3$
- $(a+b+c)(a^2+b^2+c^2-ab-bc-ca)=a^3+b^3+c^3-3abc$
- **If** $a+b+c=0$**, then** $a^3+b^3+c^3=3abc$.
- $(x+a)(x+b)(x+c)=x^3+(a+b+c)x^2+(ab+bc+ca)x+abc$

Example 6.1 Without cubing, evaluate $(28)^3+(-15)^3+(-13)^3$.

SOLUTION

$28-15-13=0$, so the sum is $3\times 28\times(-15)\times(-13)$.

16380

Example 6.2 If $a+b+c=9$, $ab+bc+ca=26$ and $abc=24$, find $a^3+b^3+c^3$.

SOLUTION

$a^2+b^2+c^2=81-52=29$

$a^3+b^3+c^3-72=9\times(29-26)=27$

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7 Common mistakes

- Writing $(a+b)^2 = a^2 + b^2$ or $(a+b)^3 = a^3 + b^3$.
- Losing the sign of the middle term in $(a-b)^2$.
- Forgetting the factor 2 in the product terms of $(a+b+c)^2$.
- Dropping the $\pm$ when taking a square root, as in finding $a-b$ from $(a-b)^2$.
- Using $a^2 + \frac{1}{a^2} = \left(a + \frac{1}{a}\right)^2 + 2$. The sign is minus.

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